

Lesson 1: Markets from Zero

Analyst onboarding · Lesson 1 of 10 · L/S Gamma Desk, QUANTT

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Goal: understand what we trade and the handful of ideas every later lesson builds on.

You need: no finance background. If you can read a percentage, you're ready.

1 Stocks, indexes and SPY

- **A stock (or share)** is a small piece of ownership in a company. If you own 1 share of a company with 1 billion shares, you own one billionth of it. Its **price** is set by buyers and sellers trading it on an **exchange** (a marketplace such as the New York Stock Exchange).
- **An index** is a scoreboard that tracks a group of stocks. The **S&P 500** tracks 500 of the largest US companies. When people say “the market went up 1% today”, they usually mean the S&P 500.
- **An ETF** (exchange-traded fund) is a fund that holds a basket of stocks and trades like a single stock. **SPY** is an ETF that holds all 500 S&P 500 companies. One SPY share costs about \$760 today, and its price moves with the whole US market.

⇒ Our desk trades **SPY** and **options on SPY** (Lesson 2). SPY is one of the most traded products in the world, which keeps trading costs low.

2 Buying, selling and “going short”

- **Long:** you own something and profit if its price **rises**. Buy 10 SPY at \$760 and sell at \$770: you make $10 \times \$10 = \100 .
- **Short:** you profit if the price **falls**. You borrow shares from your broker, sell them now, and buy them back later to return them. Short 10 SPY at \$760 and buy back at \$750: you make \$100. If it rises instead, you lose.
- **Broker:** the company that sends your orders to the exchange and holds your account. We use **Interactive Brokers (IBKR)**, in a **paper account**: a practice account with fake money but real market prices.

3 Prices have two sides: bid and ask

At any moment there are two prices:

- **Bid:** the highest price someone is willing to **pay**. You sell at the bid.
- **Ask:** the lowest price someone is willing to **sell** for. You buy at the ask.
- **Mid:** halfway between the two. **Spread:** the gap between them.

If SPY is bid \$759.99 and ask \$760.01, buying and immediately selling costs you the \$0.02 spread. That is a **transaction cost**. Every trade we make pays some of it, so a strategy that trades too often can lose money even if its ideas are right.

4 Returns: measuring gains in percent

A **return** is the percentage change in price. If SPY goes from 760 to 767.60, the return is +1%.

Quants (quantitative traders) often use **log returns**, $r = \ln(P_{today}/P_{yesterday})$. For small moves they are almost identical to percent returns (+1% vs +0.995%), and they add up neatly over many days. Our code uses them.

5 Risk and volatility: how much prices wiggle

Two stocks can both average +10% a year, but one moves $\pm 0.3\%$ a day and the other $\pm 3\%$ a day. The second is far riskier. We measure that wiggle with **volatility**:

1. Take daily returns over some period.
2. Compute their **standard deviation**: the typical size of a move away from the average. Roughly, about two thirds of days land within one standard deviation.
3. **Annualize** it by multiplying by $\sqrt{252}$, because there are about 252 trading days in a year.

Example. If SPY's daily returns have a standard deviation of 0.88%, annual volatility is $\sigma = 0.88\% \times \sqrt{252} \approx 14\%$. Going the other way, a 14% volatility means a "typical" day moves about 0.88%, or about \$6.70 on a \$760 SPY.

$$\sigma_{daily} = \frac{\sigma_{annual}}{\sqrt{252}}$$

$\Rightarrow$ Volatility, written σ (**sigma**), is the single most important number in this course. It is also the thing we trade.

6 Two kinds of volatility

- **Realized volatility (RV)**: how much SPY **actually moved** over a period. You measure it from past prices.
- **Implied volatility (IV)**: how much movement the **options market is pricing in** for the future (Lesson 3).

Our entire strategy is a bet on the gap between these two: will SPY move **more or less** than the market expects?

7 Volatility behaves in patterns

1. **Clustering**: calm days follow calm days, and wild days follow wild days.
2. **Mean reversion**: very high or very low volatility tends to drift back toward normal levels.
3. **The leverage effect**: volatility usually jumps when markets **fall**, not when they rise. Crashes are fast, rallies are slow.

These patterns are why volatility can be **forecast** (Lesson 7), unlike the direction of the market, which is very hard to predict.

8 What our desk does, in one paragraph

We are the **L/S Gamma desk** at QUANTT. Every day we forecast how much SPY will move over the next month and compare that with how much movement options are priced for. If options look **too cheap**, we buy them. If they look **too expensive**, we sell them. Then we remove the direction bet by trading SPY shares (Lesson 5), so we only make or lose money on **how much** SPY moves, not **which way**.

9 Words to know

- **SPY:** ETF tracking the S&P 500. **Long / short:** profit from up / down moves.
- **Bid / ask / mid / spread:** buy price, sell price, the middle, and the gap.
- **Return:** percent change. **Volatility (σ):** annualized standard deviation of returns.
- **RV / IV:** volatility that happened / volatility the options market expects.
- **Paper account:** practice brokerage account with fake money.

10 Check yourself

1. You short 5 SPY at \$760 and buy them back at \$766. What is your profit or loss?
2. SPY's annual volatility is 20%. Roughly how big is a typical daily move, in percent?
3. Why does trading more often cost more, even at fair prices?

Answers: (1) $-5 \times \$6 = -\30 . (2) $\sigma_{daily} = 20\% / \sqrt{252} \approx 1.26\%$. (3) Each trade pays part of the bid-ask spread.